CLICS — Prototype Summary (for non-technical readers)

CLICS = *Clinical Learning & Intelligence Credit System* A pilot platform for a Thai medical licensing body, July–September 2026.

Live links

Physician portal	https://physician-ui.vercel.app
Audit dashboard (Admin / Auditor / Superadmin)	https://audit-dashboard-rouge.vercel.app
Backend API	https://backend-production-c12e4.up.railway.app

> **To review the pilot:** sign in to the **audit dashboard** as **Superadmin** (sees everything) or **Admin** (sees the cohort + onboarding history). Logins are in `LOGIN-ACCOUNTS.txt` and summarised in §10 below.

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1. What problem does it solve?

Today, doctors earn continuing-education (CME/CPD) credit largely by **attending** lectures — which proves they showed up, not that they learned. CLICS changes the basis of credit from *attendance* to **the quality of a doctor's clinical thinking.**

A doctor has a real clinical conversation with an AI tutor about a patient case. The system then scores **how well the doctor reasoned** — the questions they asked, how they weighed evidence, whether they challenged the AI — and awards credit on that basis.

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2. Who uses it? (four roles)

Role	In plain terms	What they do
Physician (แพทย์อาสา)	The doctor earning credit	Has AI case discussions, gets scored, tracks progress
Admin	The pilot coordinator	Approves new members, oversees cases, sees statistics + onboarding history
Expert Auditor	A senior doctor checking quality	Independently re-scores cases "blind" to confirm the AI is fair
Superadmin	The system owner	Manages all accounts + role capacity, oversees the whole programme

3. What does a doctor actually experience?

1. **Get an account** — request access (or sign in), then an admin approves them into the programme.
2. **Log in** to the physician portal.
3. **Start a case** ("Clinical Learning Episode") and type the patient's main problem.
4. **Discuss it with the AI tutor** — ask questions, propose a diagnosis, think it through. The AI is a **mentor, not an answer machine**: it explains its reasoning, **backs claims with references**, **politely corrects mistakes** instead of just agreeing, and **ends every reply with a question** that pushes the doctor to think one level deeper.
5. **Finish the session.** The system checks it was a genuine learning episode, then scores the doctor's reasoning.
6. **See the result** — a 7-part score and a personal progress chart over time.

Doctors can also **organise their conversations into folders**, name/rename each one, and move them around — all on a tidy dashboard.

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4. How is "quality of thinking" measured?

- **PIS-7** — every session is scored on **7 dimensions** of clinical reasoning (quality of questions, depth of reasoning, use of evidence, self-reflection, and more), shown as an easy-to-read radar chart.
- **Fairness check (ICC)** — human expert auditors re-score a sample of cases *without seeing the AI's score*. We then measure how closely the AI and the human experts agree. The programme only proceeds to full scale once that agreement is high enough (target **0.75**). This is what makes the AI scoring **trustworthy**, not a black box.

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5. Who can join, and how (new: membership & access)

The platform now manages a **real cohort safely**:

- **Request access** — a clinician applies from the portal (by email, or Google/LINE sign-in). The request waits as **pending**.
- **Approval into a capped slot** — a superadmin/admin approves them into an available place for their role (each role has a set capacity). On approval they get a stable **pseudonym** (e.g. *P-001*) — the only identifier used in any review or export.
- **Guest mode** — a curious clinician can **"Try as guest"**: 2 AI prompts, then it ends, with **nothing saved at all**.
- **Private by design** — real names/emails (incl. social-login identity) are kept in an **encrypted, on-premise vault** and never reach the cloud AI; only the pseudonym is ever visible downstream.

6. The 1-week pilot dry run (soak test) — results

Before opening to real users, we ran a **full simulated week on the live system** to prove the whole lifecycle and stress-test the fairness check.

Setup (all on a clean database, fully traceable by superadmin/admin):

- **16 accounts** onboarded through the real flows: 1 superadmin, 2 admins, **10 physicians** (spanning weak → excellent reasoning), 3 expert auditors.
- The 10 physicians ran **15 case sessions** with the real AI tutor over a backdated week (May 31 – Jun 6); the 3 auditors then blind-scored the credit-eligible sessions.

What we learned:

- ✅ **The whole lifecycle works end-to-end** — onboarding → AI case discussion → quality gate → 7-part scoring → progress charts → auditor blind-scoring → fairness check.
- ✅ **The quality gate behaves as designed** — weak/short sessions were auto-rejected, average sessions flagged for manual review, only genuine reflective sessions reached scoring (10 of 15).
- ⭐ **The fairness check did its job — the key finding.** The **three auditors agreed with each other almost perfectly (ICC 0.97)**, but the **AI's scores agreed with the human experts only at 0.66 — below the 0.75 bar.** The AI is currently a touch **lenient**, over-rewarding well-structured-but-shallow reasoning. **This is exactly what the safeguard exists to catch:** it tells us to **calibrate the AI against expert judgement** at the **July calibration workshop** before any credit is issued.
- ✅ **Reliability hardened** — the dry run surfaced a case where a brief network hiccup could leave a scored session unrecorded; this was **fixed and re-deployed** during the review (scores now retry and self-recover).

A full written review (with the funnel, per-doctor outcomes and the agreement numbers) is in `docs/SOAK-REVIEW-2026-06.md`.

7. Scale & reliability — can it handle many users at once?

We load-tested the **live** system to confirm it holds up for a real cohort:

- **100+ concurrent users — confirmed.** Everyday use (logging in, dashboards, viewing scores/sessions) ran cleanly at **200 simultaneous requests** and **110 distinct logged-in users** with **zero errors** — comfortably past the 100-user goal.
- **AI tutor under heavy load — hardened.** When many doctors chat with the AI at the *exact same moment*, success at a 40-at-once burst was raised from **33%** → **88%** by adding smart **queuing + automatic retry**. Realistic usage (doctors don't all send at the same instant) is effectively **100%**.
- **One remaining lever is a plan upgrade, not code.** Live diagnostics pinpointed the limit precisely — the AI provider's **per-minute output allowance** on our current plan. Raising that

tier is the single change needed to handle very large simultaneous AI bursts.

- **Cost-controlled** — all load testing stayed within a tight, pre-agreed API budget.

A full written report (numbers, methods, recommendations) is in `docs/LOADTEST-2026-06.md`.

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8. Privacy & safety (important for the pilot)

- **Patient/doctor identifying information stays on-premise.** Only **de-identified** text is ever sent to the cloud AI.
- Real identities (incl. Google/LINE) live in an **encrypted on-premise vault**; only pseudonyms (*P-001…*) appear in reviews/exports. Guest sessions store nothing.
- The AI is built to **avoid making things up** and to **cite evidence** — reducing the risk of confident-but-wrong advice.
- All accounts are role-restricted: each person only sees what their role allows.
- **The superadmin (system owner) account is locked** — it can never be disabled, demoted, deleted, or duplicated, *even by itself*, enforced in both the app and the database (always exactly one superadmin).

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9. What is *not* in this prototype yet (roadmap, not bugs)

- **Calibrating the AI scorer** to the expert consensus — the July workshop's main task (see §6).
- Rotating between **multiple AI providers** (today it runs on a single, high-quality model).
- **Raising the AI provider's per-minute output limit** — to support very large *simultaneous* AI-chat bursts (the binding limit was pinpointed precisely in load testing; see §7). A plan/billing change, not code.
- **Full pseudonymisation** of all free-text identifiers (today: an encrypted identity vault + pattern-based de-identification).
- The automated **credit-issuing engine** (scores are produced; formal credit issuance is a later step).
- Real **Google/LINE sign-in** handshake (the buttons + the private identity vault are built; the live social-login handshake is stubbed pending credentials).

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10. How to try it

Live links are at the top of this document. Full credentials (all roles) are in `LOGIN-ACCOUNTS.txt`. For a manager review, the two accounts to use are:

Role	Email	Password	What you'll see
Superadmin	`superadmin@clics.th`	`ClicsSuper#2026r2`	Everything — all 17 members, role-slot pools, every session, the fairness (ICC) numbers
Admin	`admin@clics.th`	`ClicsAdmin#2026r2`	The cohort's sessions, the manual-review queue, and the **new onboarding history** (who was approved, by whom, when)

Both sign in at the **audit dashboard** (`/login`). `LOGIN-ACCOUNTS.txt` also lists a sample physician and auditor login from the live dry-run cohort if you want to see those consoles.

These are throwaway demo passwords for the walkthrough and will be replaced before the real pilot.

> This is a **working prototype** for hands-on evaluation — built to demonstrate the concept end-to-end, not the final production system.